Table of Contents

CycleProDefTopic

<	>

About CyclePro

About CyclePro

CyclePro is a tool that is used with the UltraSerial10G to display PCIe PA protocol packet information. It lets you analyze the data using search and annotation features. CyclePro can take data from an executing program. It can also save the data to a file for later redisplay and analysis.

For more information, see Using CyclePro.

For more information on the UltraSerial10G, see About the UltraSerial10G.

IG-XL includes additional Protocol Aware tools:

- Protocol Definition Editor:
  - Define a protocol from scratch
  - Edit an existing protocol definition
  - Graphically view frame waveforms
- Protocol Studio:
  - Displays Protocol Aware (PA) frames at various levels of abstraction
  - Enables programming of PA port settings
  - Enables initiation of frames

For more information, see:

- About Protocol Definition Editor
- About Protocol Studio

For information about Protocol Aware in general and nWire in particular, see the following:

- About Protocol Aware
- nWire Protocol Aware

<	>

CycleProUsing.2.01

<	>

Using CyclePro

Using CyclePro

This section describes using CyclePro to analyze data. The following sections are available:

- CyclePro Window
- Getting Data
- Using Search and Annotate to Analyze Data

To bring up Cycle Pro, on the IG-XL Tools tab, in the Launch group, select Tools>Cycle Pro.

<	>

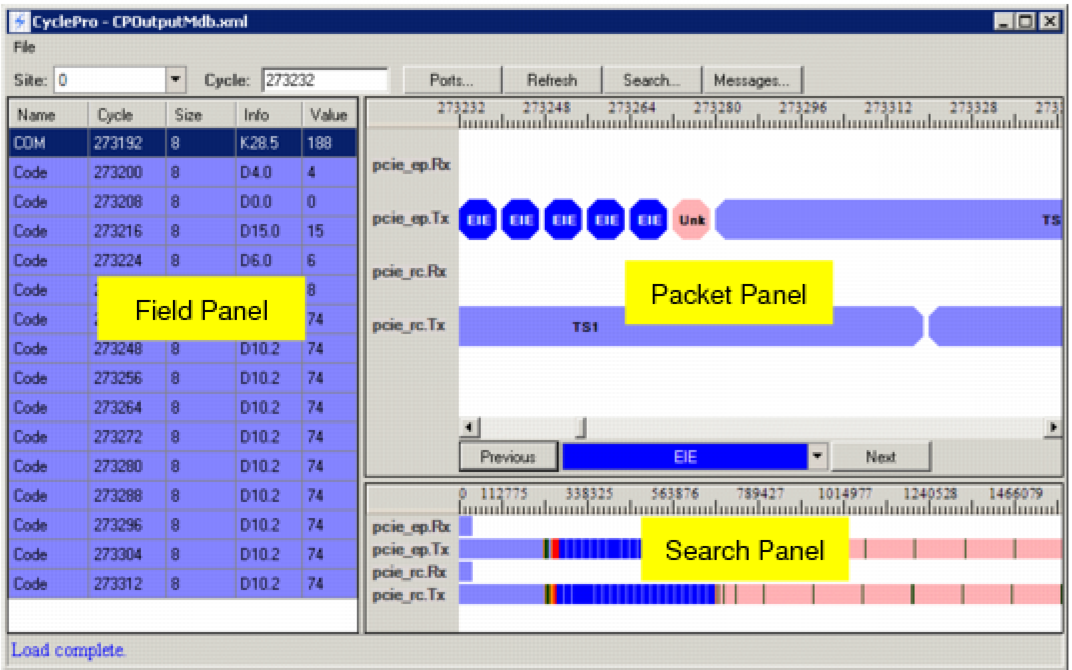
CycleProUsing.2.02

<	>

Using CyclePro : CyclePro Window

CyclePro Window

When data has been loaded and analyzed, the window looks like this:



CyclePro Window, Data Loaded and Analyzed

The parts of the window are as follows:

Packet Panel	Displays a zoomed-in view of packet traffic on a per-link, per-cycle basis. “Ticks” on the ruler at the top of the panel represent cycles.
Field Panel	Displays field-level information for the currently selected packet.
Search Panel	Shows all packet traffic on a per-link, per-cycle basis. It lets you select the part of the complete data to show in the Packet Panel and Field Panel. You can zoom in and out on the Search Panel data.

	"Ticks" on the ruler at the top of the panel represent cycles.
Cycle:	This control at the top of the window displays the cycle currently selected on the bars in the Search Panel.

<	>

CycleProUsing.2.03

<	>

Using CyclePro : Getting Data

Getting Data

You can get data from an executing job or from a file. The following topics are included:

- From an Executing Job
- From a File
- Selecting Site and Cycle

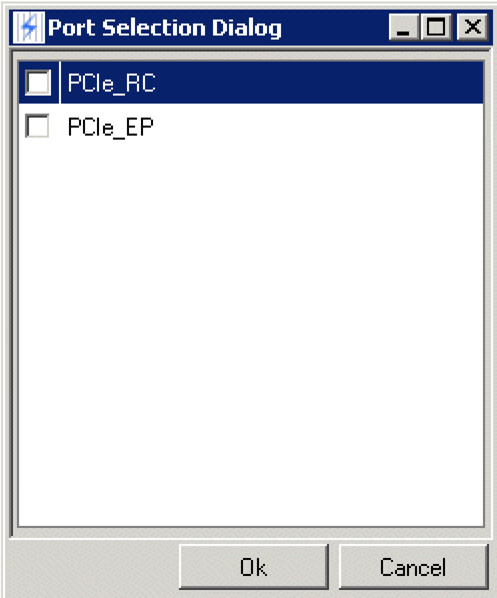
From an Executing Job

You can get data from an executing job. You must set up CMEM to be in cycle mode (not payload), and you must set the capture size. Once you have some transaction data in CMEM, you can load the data. If you put data into CMEM only once, you can wait until the program is finished. If you put data into CMEM multiple times, you must break at the point where you have the data you want to view.

To load data from the executing job:

1. At the top of the CyclePro window, click the Ports button.

The Port Selection Dialog window appears, listing the ports defined in the job:

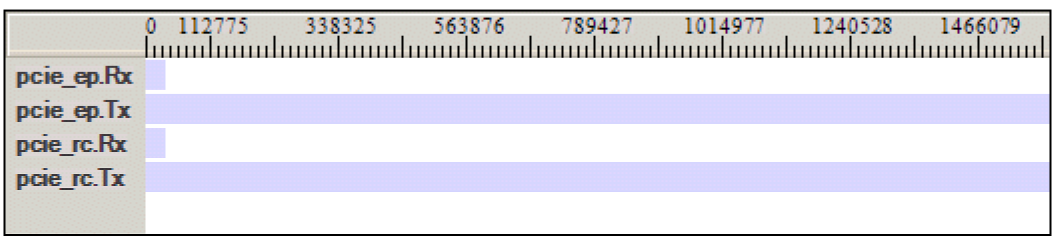


Port Selection Dialog Window

2. Check the boxes for the ports whose data you want.

3. At the top of the CyclePro window, click the Refresh button to get the data for those ports.

If you have selected both PCIe ports shown in the window above, the data in the Search window looks like this:

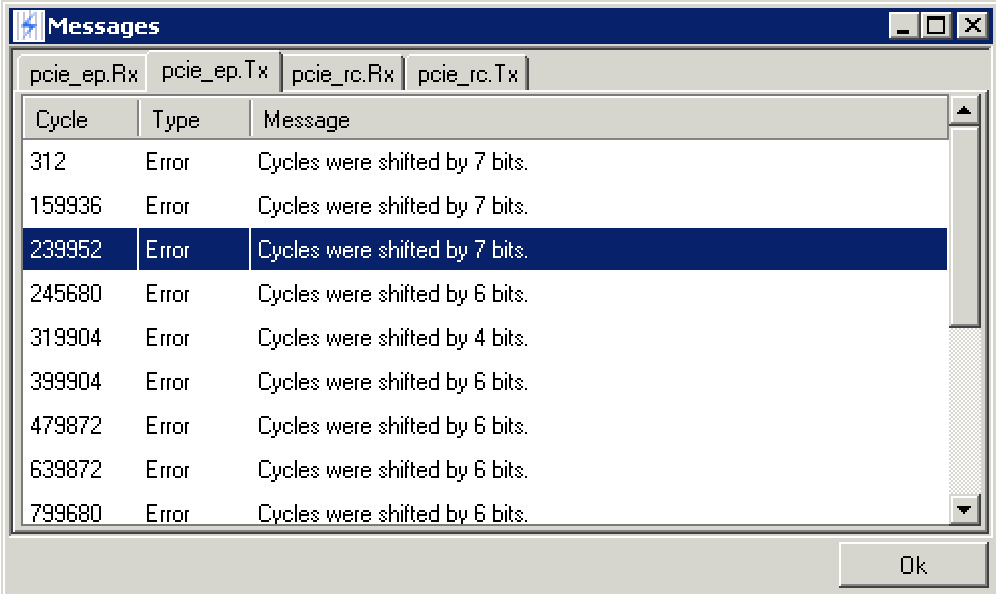


Search Window with Data

In this example, Rx and Tx links found for each port:

- Tx data comes from CMEM
- Rx data comes from HRAM

As CyclePro loads the data, it translates it into a format that it can use. As it does so, it writes messages about problems with formatting. To see the messages, click this button to bring up the Messages window:



Messages Window

The window has one tab per link. The messages are about the displayed data generated when the raw ones and zeros are translated to PCIe form. The window displays the cycle, message type, and message. Clicking a particular message takes you to the relevant cycle in the data that the message is related to.

From a File

You can also save data from a CyclePro session to load and analyze it later:

- To save the current data to a file, use File>Save.

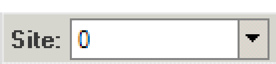
- To open a saved file, use File>Open.

Both commands bring up the browser.

The saved file has the .xml extension.

Selecting Site and Cycle

Once you load the data, you can use controls to select the data to be shown:



The Site control lets you select the site whose data is to be displayed.

The Cycle control displays the current cycle number, which is defined as the leftmost cycle tick of the Packet Panel. You can enter a number to go to that cycle: the display changes as you enter additional numbers. If you enter a number that is higher than the last cycle, you go to the end of the display.

<	>

CycleProUsing.2.04

<	>

Using CyclePro : Using Search and Annotate to Analyze Data

Using Search and Annotate to Analyze Data

This section is a step-by-step example of using Cycle to analyze data. The steps are as follows:

- Annotating Field Data

<	>

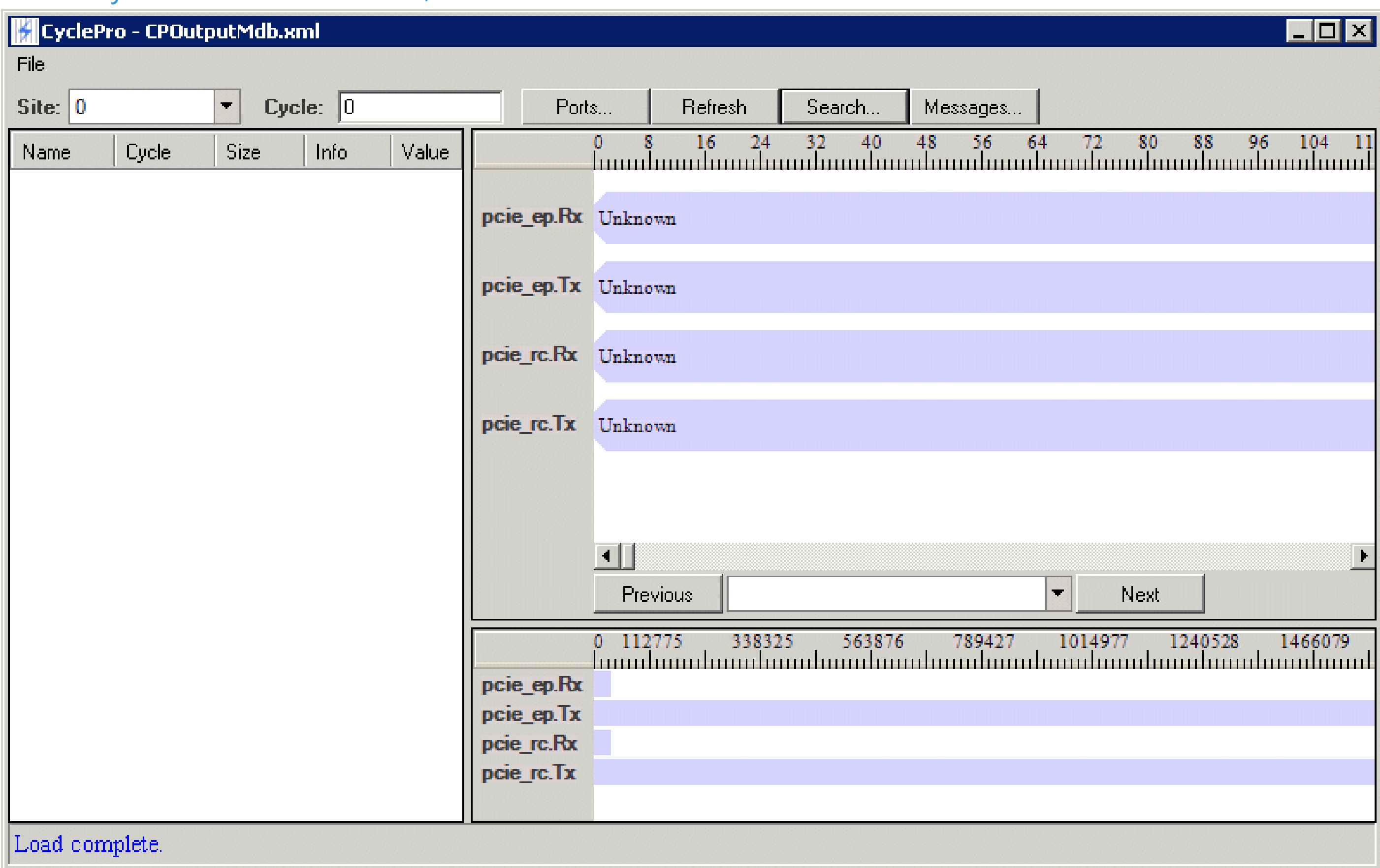
CycleProUsing.2.05

<	>

Using CyclePro : Using Search and Annotate to Analyze Data : Unanalyzed Data

Unanalyzed Data

When CyclePro first loads data, it looks like this:



CyclePro with Data Loaded, Not Analyzed

To analyze the data, you must differentiate and label different aspects of the packets and fields. You can do this using Search to apply different colors to the different items.

You can create a search using the Search button or from the context menu in the Packet and Field Panels.

<	>

CycleProUsing.2.06

<	>

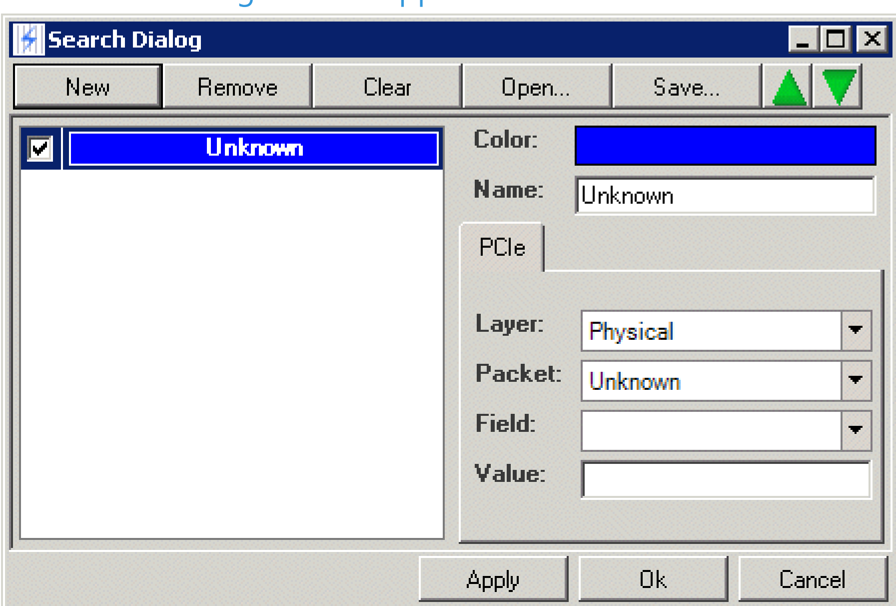
Using CyclePro : Using Search and Annotate to Analyze Data : Coloring a Packet

Coloring a Packet

As a first step in analyzing the data, you can make Unknown packets a separate color so that known packets can be found.

1.	Right-click an Unknown packet in the Packet Panel and select Create Search.

The Search Dialog window appears:

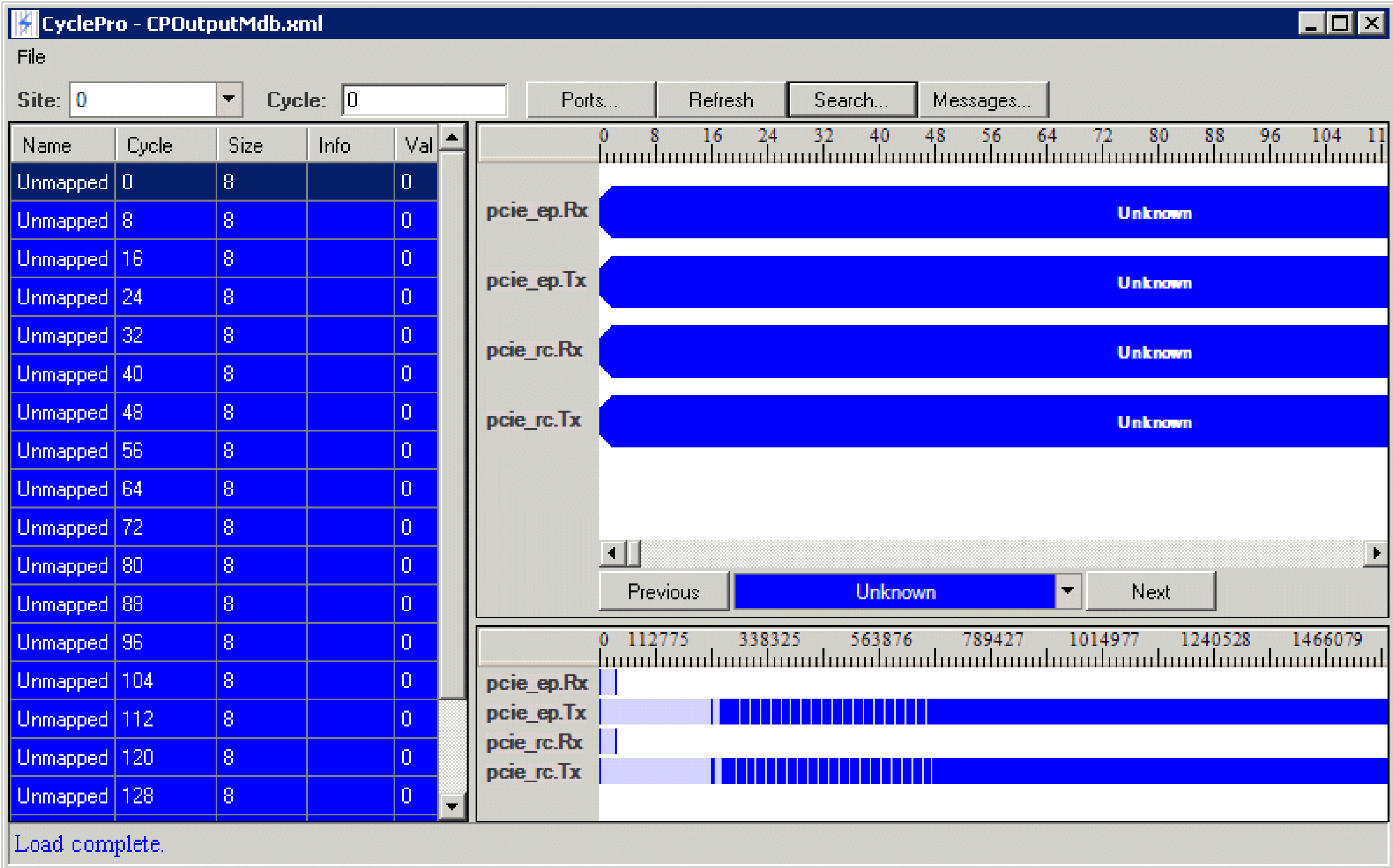


Search Dialog Window

2.	Select a color for the Unknown packets.

3.	Click Apply or Ok.

The search is applied to the data and the Packet, Field, and Search Panels are colorized using the color you selected. The following figure shows Unknown packets colored blue.



CyclePro Display, Unknown Packets Colored Blue

CycleProUsing.2.07

<	>

Using CyclePro : Using Search and Annotate to Analyze Data : Changing the Color

Changing the Color

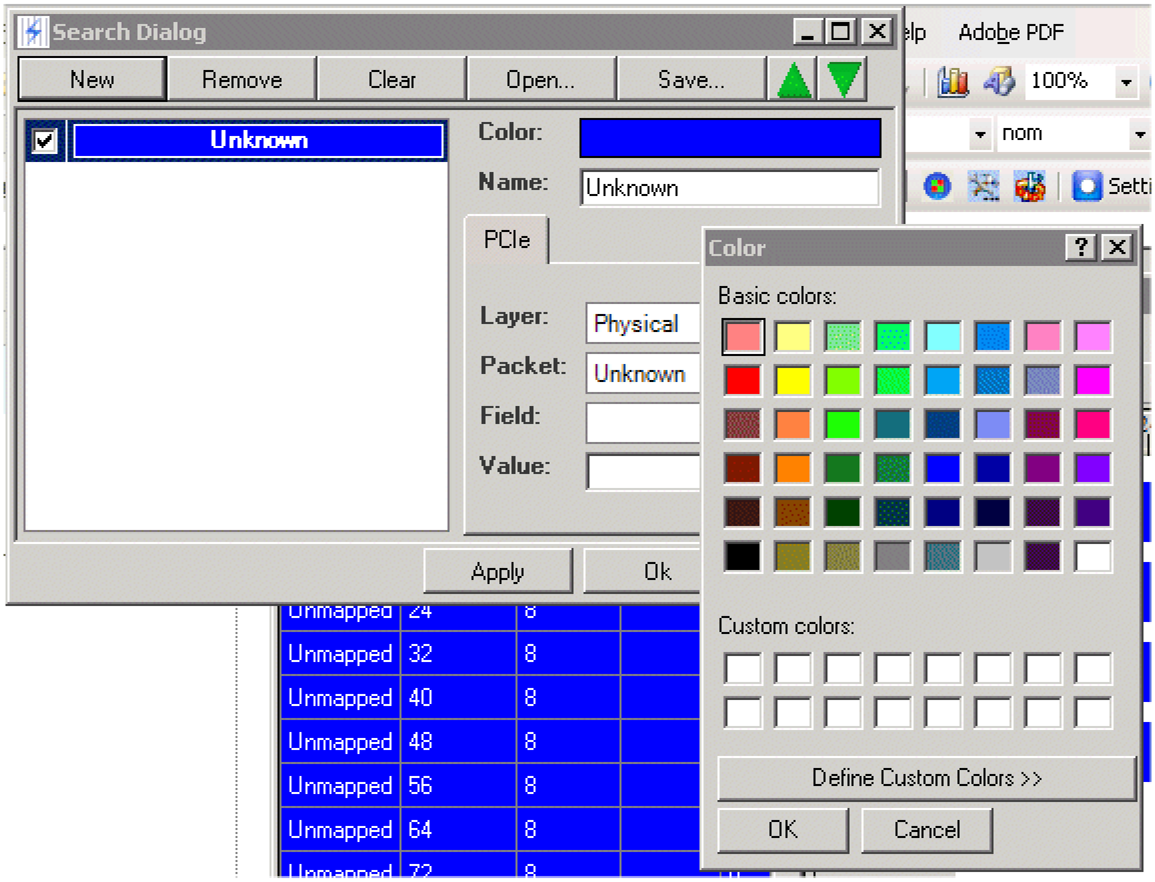
You can change the display color.

1.	Click the Search button to bring up the dialog window.

If you right-click a packet, a new search item is added to the search. Using the Search button does not add a new search.

2.	Double-click the color box to the right of the Search: label.

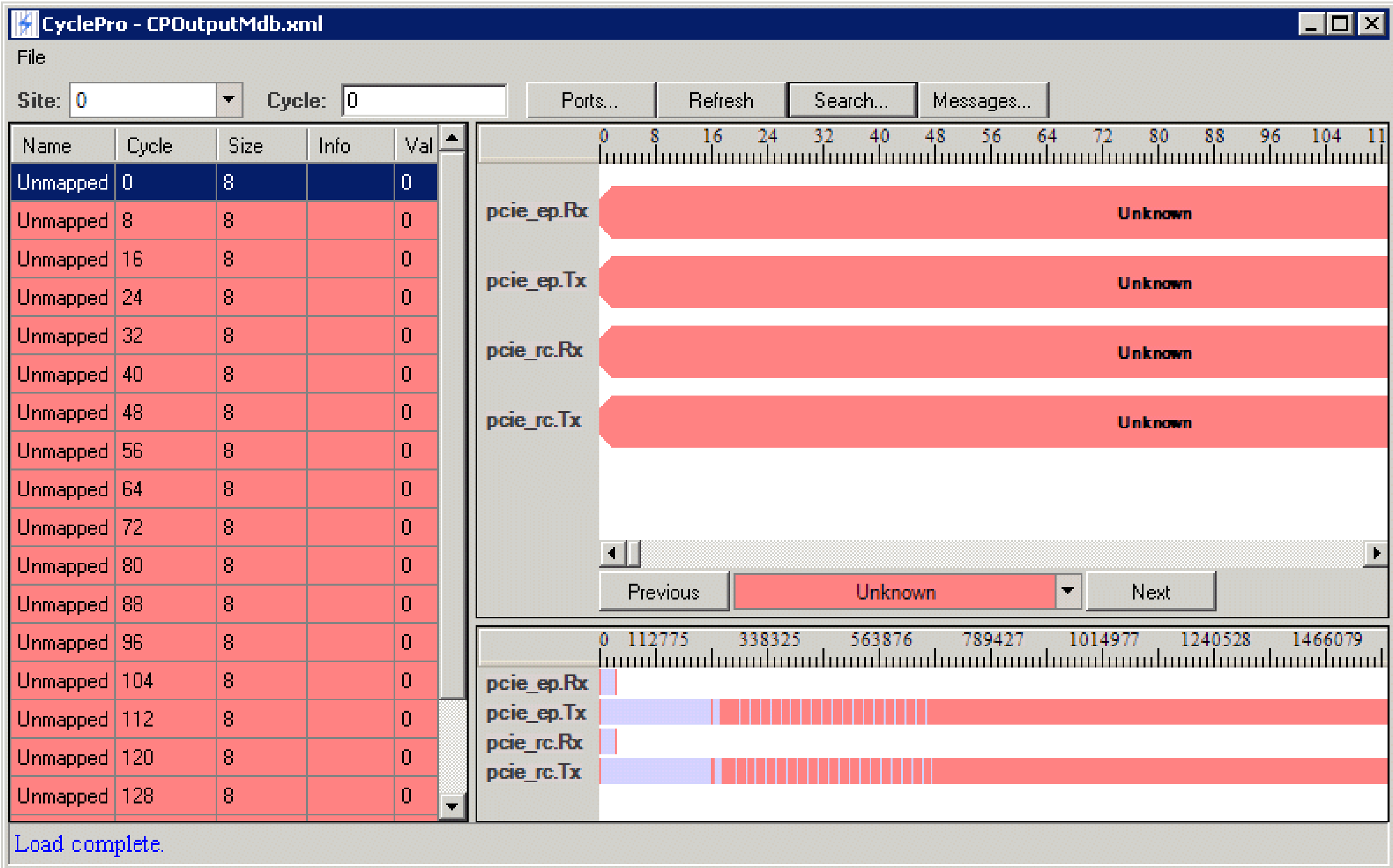
A color chooser comes up:



Search Color Chooser

3. Select the color you want and click OK on the chooser.

When you click Apply or Ok, the color is applied to the display:



CyclePro Display, Display Color Changed

<	>

CycleProUsing.2.08

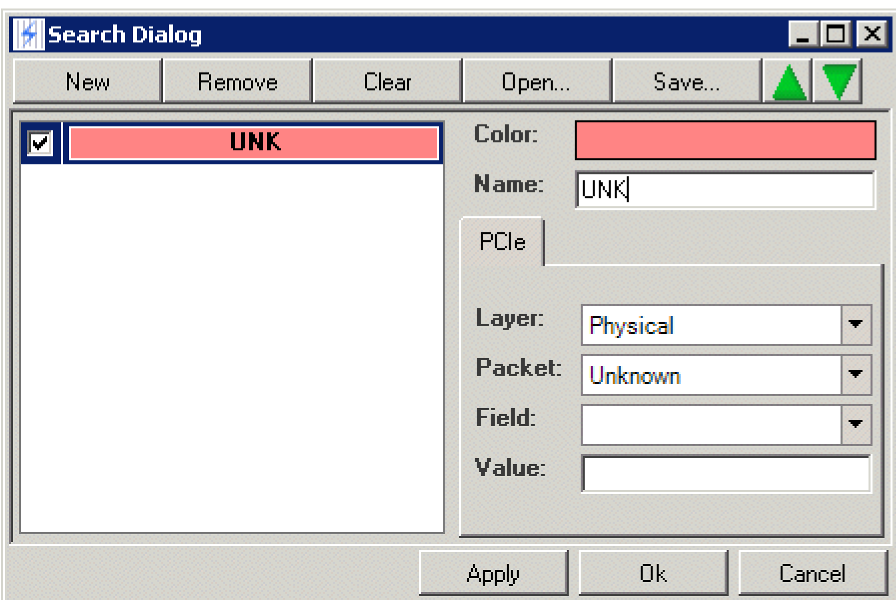
<	>

Using CyclePro : Using Search and Annotate to Analyze Data : Changing the Name

Changing the Name

You can assign a nondefault name shown for a packet. For example, you may want the abbreviation UNK.

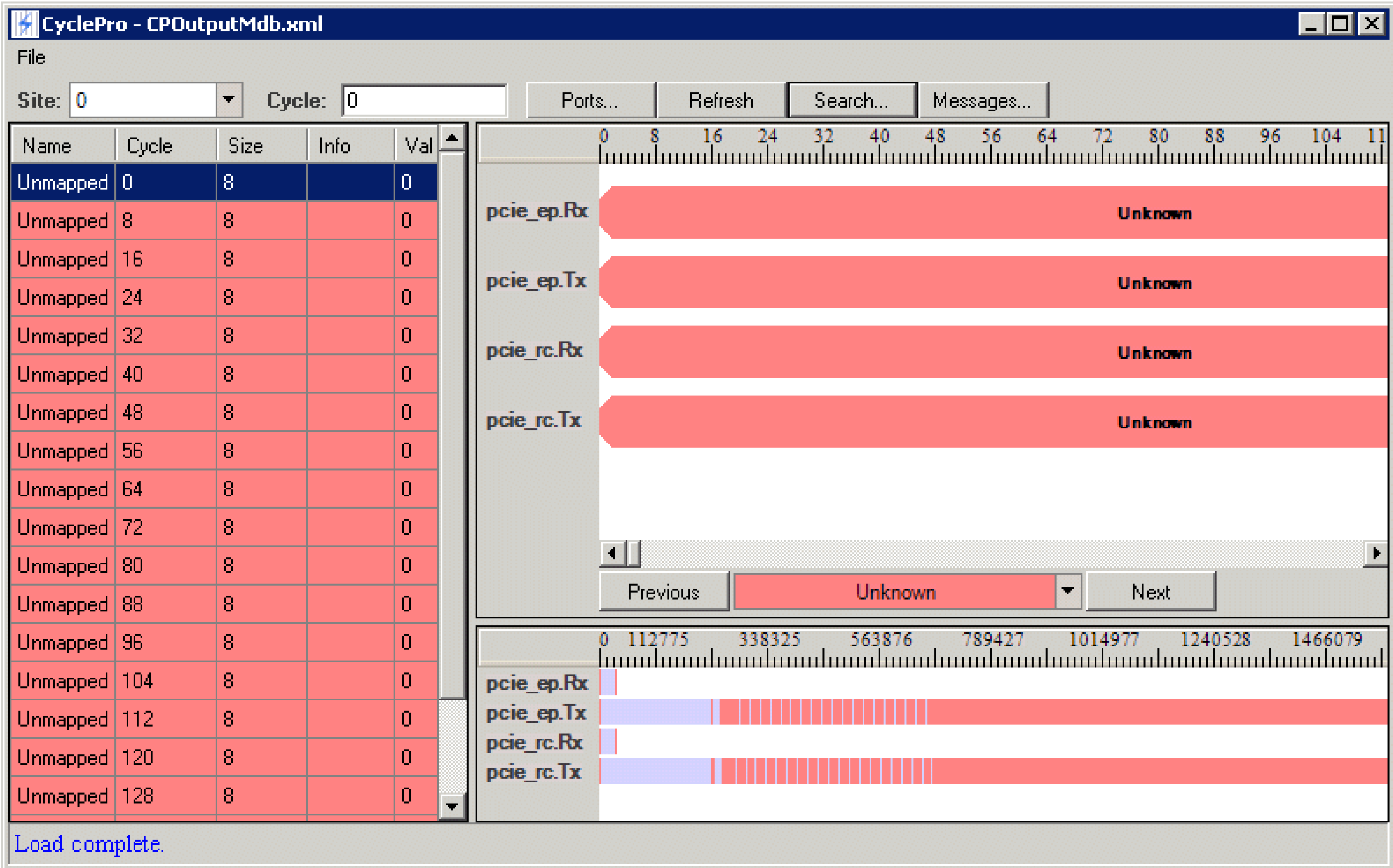
1.	Click Search to bring up the Search Dialog window.



Search Dialog Window

2.	In the Name: field, enter “UNK.”

When you click Ok, the new name is used.



Nondefault Name Assigned for a Packet

< >

CycleProUsing.2.09

<	>

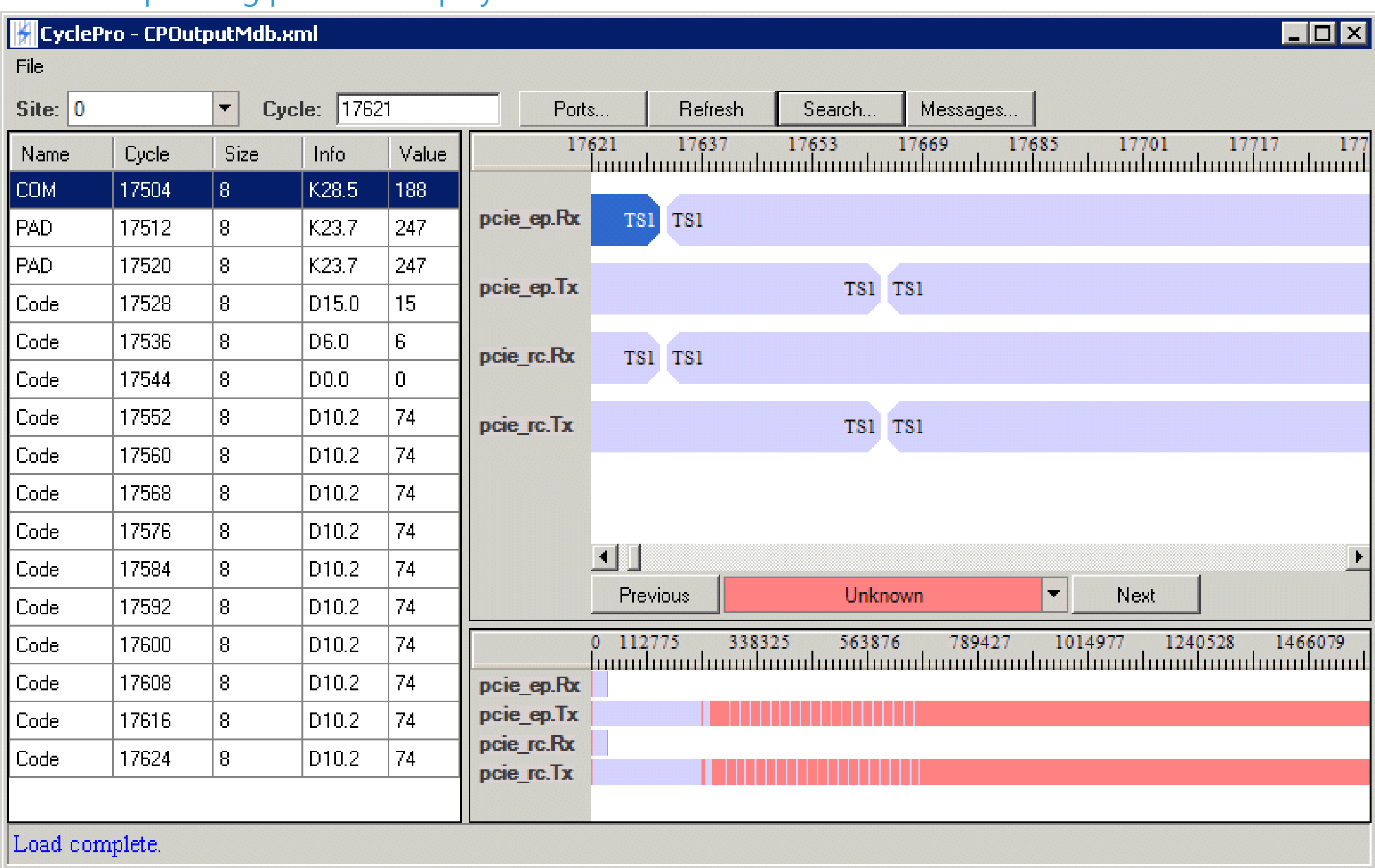
Using CyclePro : Using Search and Annotate to Analyze Data : Adding Packets

Adding Packets

Now that you have identified the Unknown packets, you can identify and color other packets.

1.	In the Search Panel, click the light blue section.

The corresponding packet is displayed in the Packet and Field Panels:

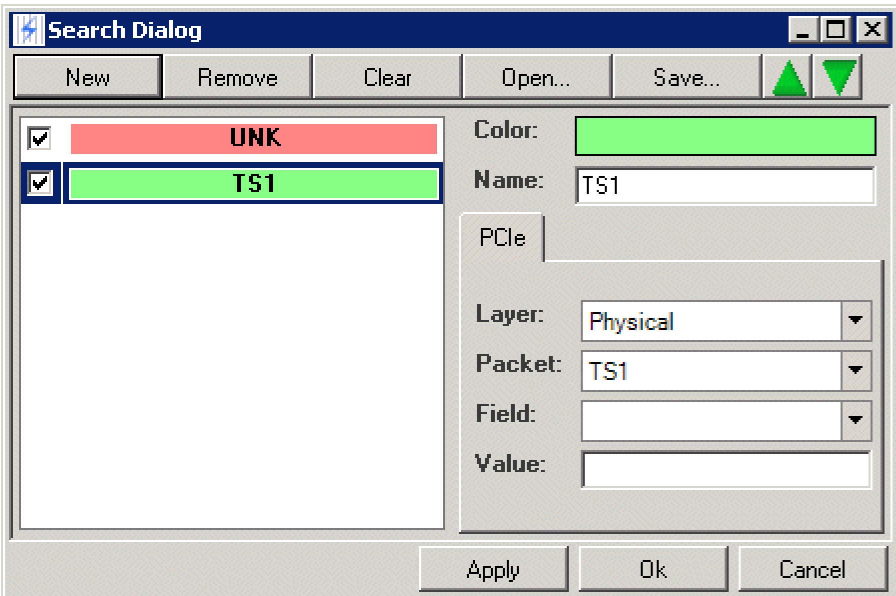


Displaying a Packet in CyclePro

2. RIght-click TS1 in the Packet Panel and select Create Search.

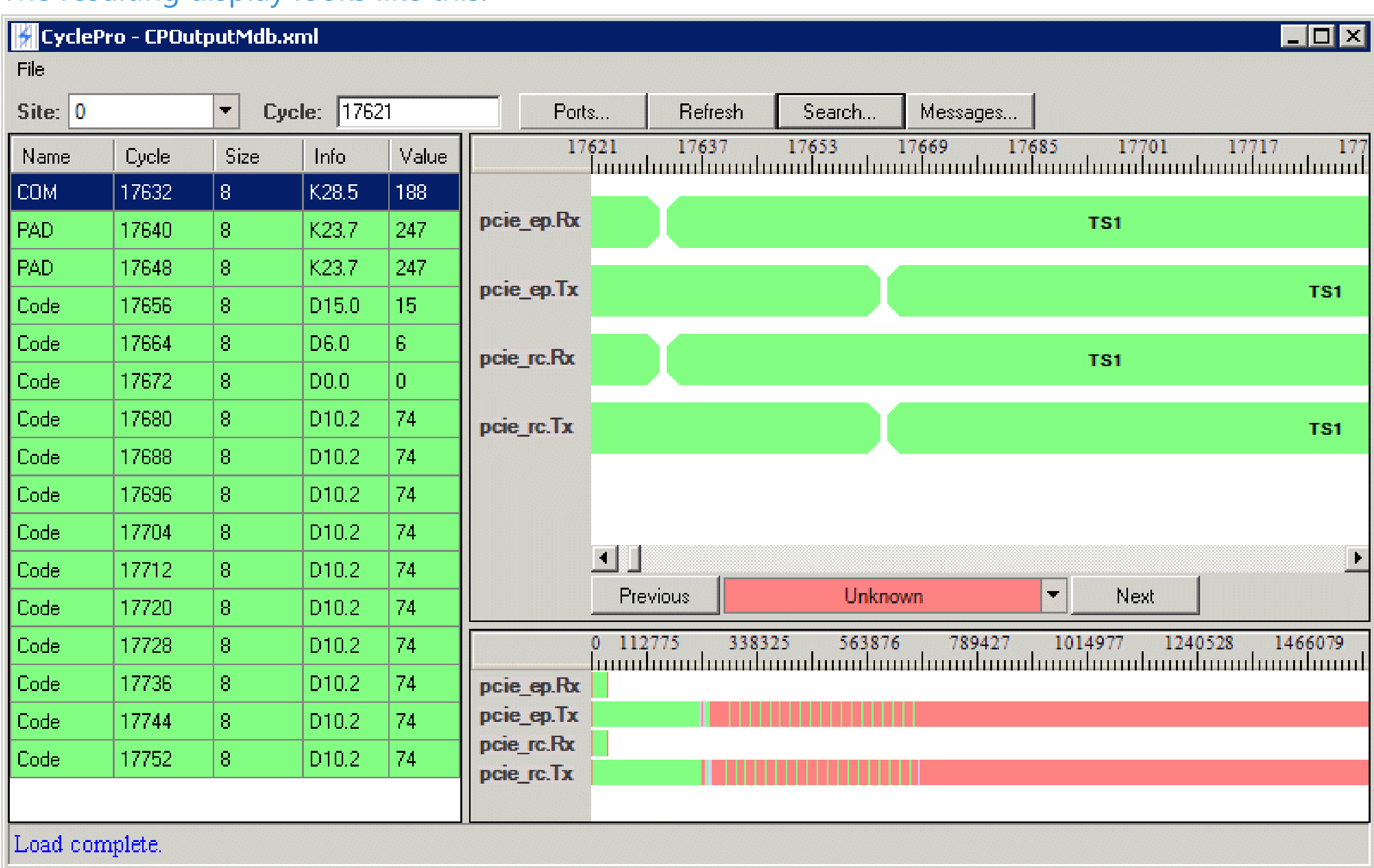
The TS1 item is now added to the dialog box.

3. Select a nondefault color: for example, light green:



Packet Added to Search with Specified Color

The resulting display looks like this:



Packet Color Specified in CyclePro

You can now follow the same steps for adding a color to the Skip packet, which is visible in the Packet Panel.

<	>

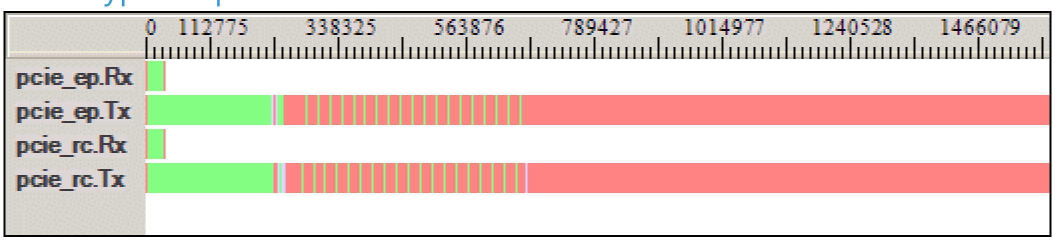
CycleProUsing.2.10

<	>

Using CyclePro : Using Search and Annotate to Analyze Data : Zooming In on the Search Panel

Zooming In on the Search Panel

Other types of packets are indicated in the Search Panel that are still the original light blue:

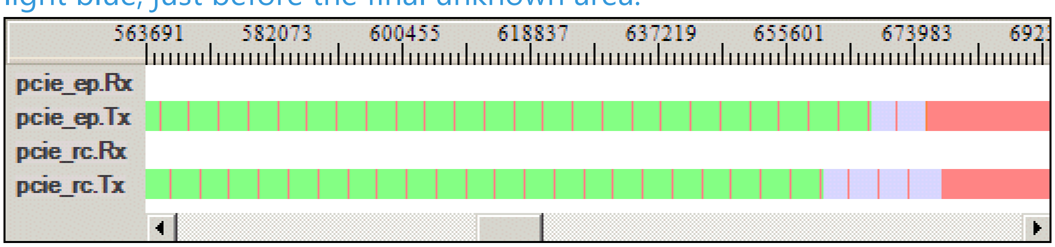


Search Panel Showing Packets

To find these, you can zoom in on a section of the Search Panel:

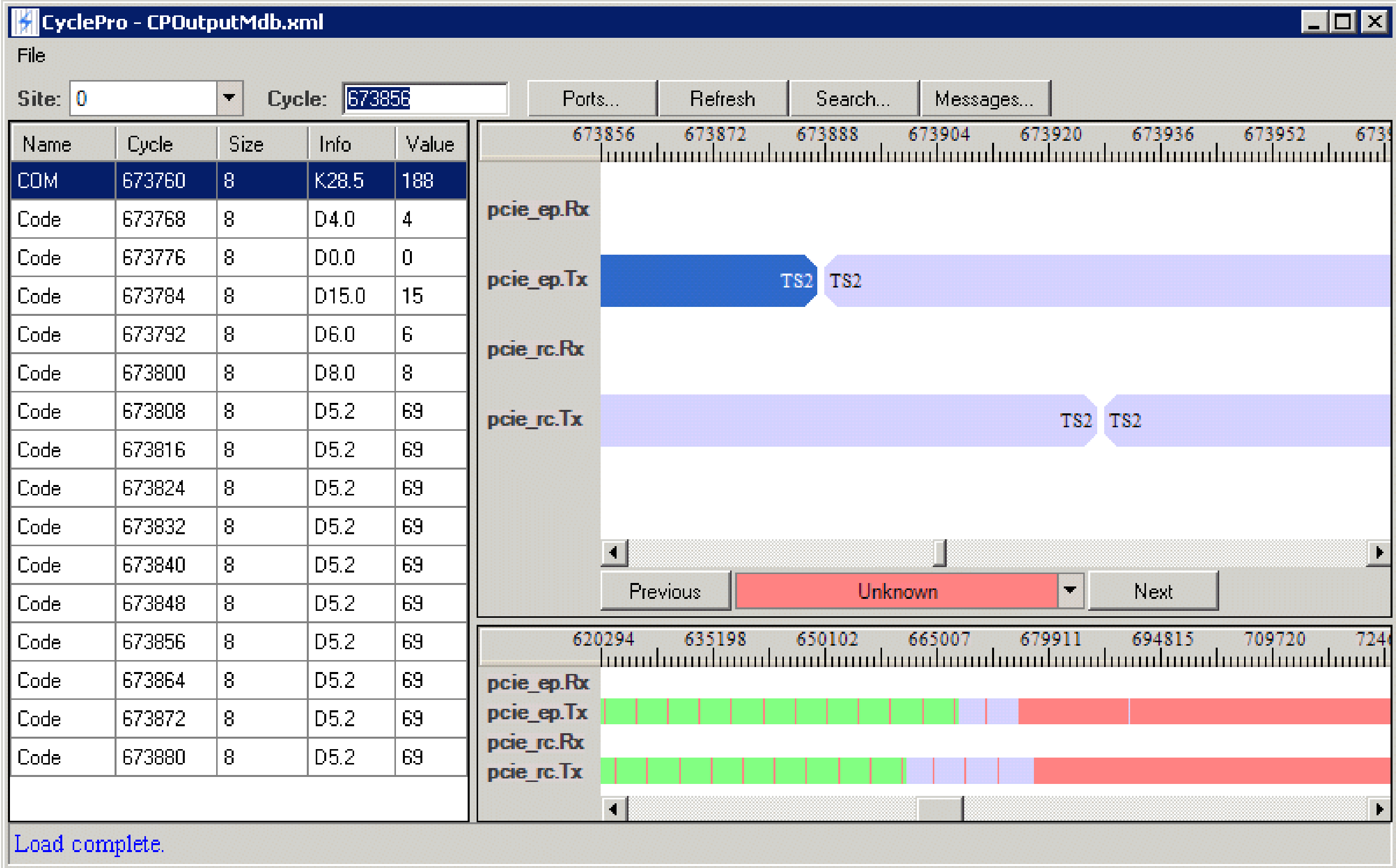
1. Left-click the ruler in the Search Packet at the point where you want to the search to begin.
2. Drag to cursor to the point where you want the zoom to end and then release.

In this example, you can zoom in on the section of either .Tx bar that has the original default color, light blue, just before the final unknown area:



Search Panel Showing Packets, Zoomed In

If you then click a light blue area in the bar, that area is displayed in the Packet Panel:



Displaying Zoomed In Area in CyclePro

You can right-click TS2 to add it to the search. Select a nondefault color, such as yellow.

To unzoom the Search Panel, left-click once anywhere in the ruler in the panel. The panel zooms out completely, showing all the data again.

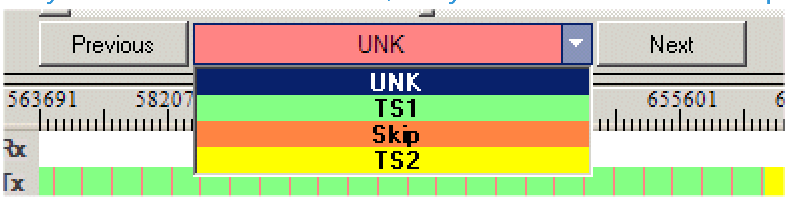
CycleProUsing.2.11

<	>

Using CyclePro : Using Search and Annotate to Analyze Data : Navigating to Items

Navigating to Items

As you add items to Search, they are added to the drop-down menu in the Packet Panel:



Search Items in Packet Panel Drop-Down Menu

You can select an item and navigate to the Previous or Next instance of it.

<	>

CycleProUsing.2.12

<	>

Using CyclePro : Using Search and Annotate to Analyze Data : Adding Field-Level Searches

Adding Field-Level Searches

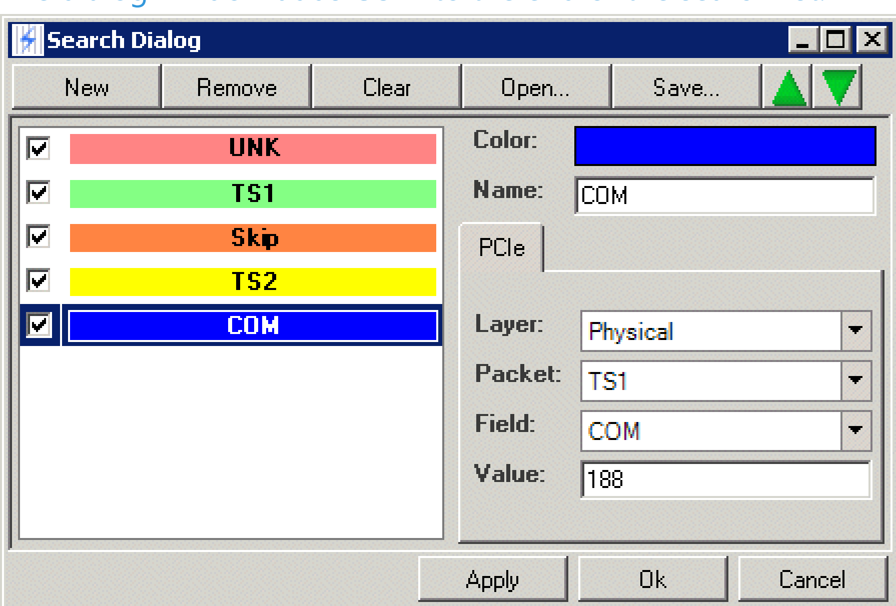
You can also add field-level search:

1. Select a TS1 packet.

The Field Panel shows fields named COM and PAD.

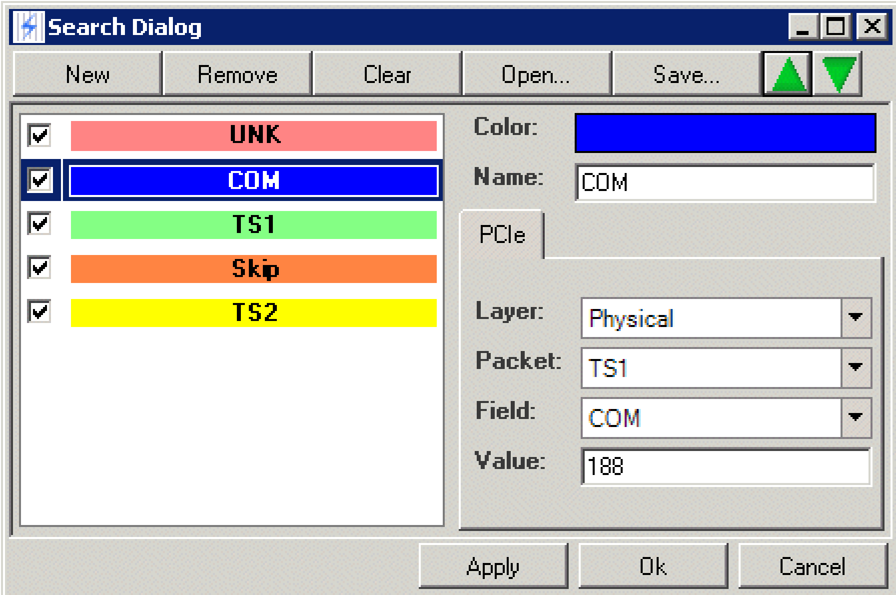
2. Select COM, right-click, and select Create Search.

The dialog window adds COM to the end of the search list:



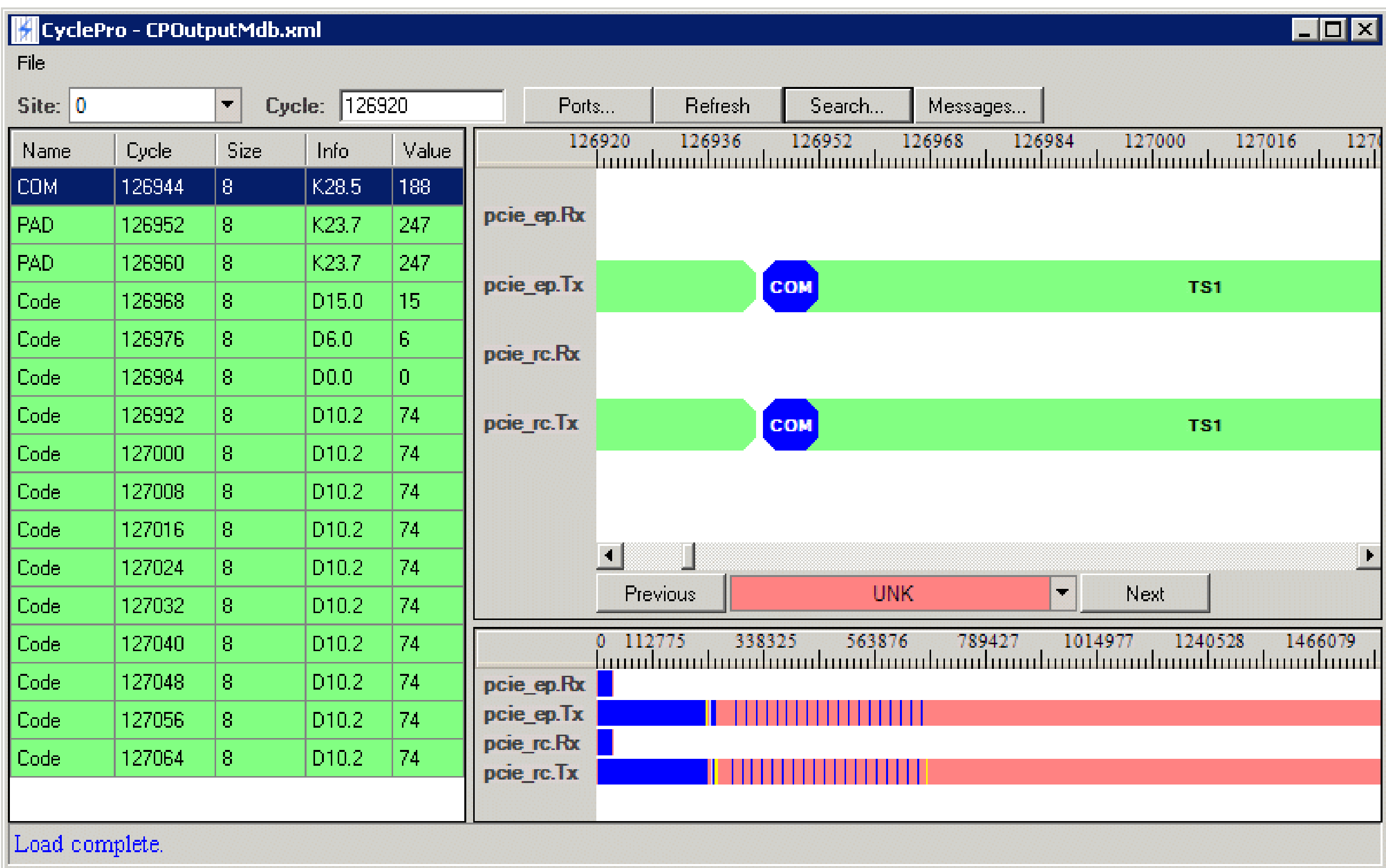
Adding Field Level Search

The order of the list is significant. If you leave COM at the end of the list, the color for TS1 is applied first and has precedence. Because COM is contained within TS1, it is not visible if the TS1 color is applied first. If you want the COM field's search to be displayed, use the up-arrow button in the upper right-hand corner of the Search Dialog to move COM's search up in priority ahead of TS1:



Changing the Priority of Search Items COM and TS1

The COM fields now appear on the TS1 transactions in the Packet Panel and are also highlighted in the Field and Search Panels:



CyclePro Display Showing COM Fields

<	>

CycleProUsing.2.13

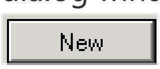
<	>

Using CyclePro : Using Search and Annotate to Analyze Data : Creating a Search in the Search Window

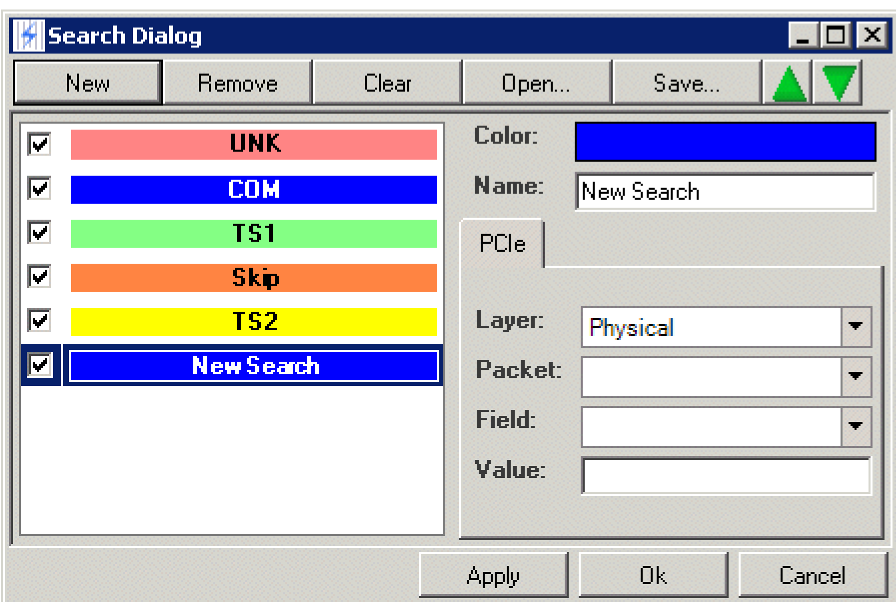
---

Creating a Search in the Search Window

The examples so far have shown creating a search by right-clicking in either the Packet or Field Panel and selecting Create Search, which automatically adds the item to the bottom of the search list of the Search Dialog. You can also click the Search button to bring up the dialog with the list as is. You can then perform certain actions on the list using the buttons and other controls on the dialog window.



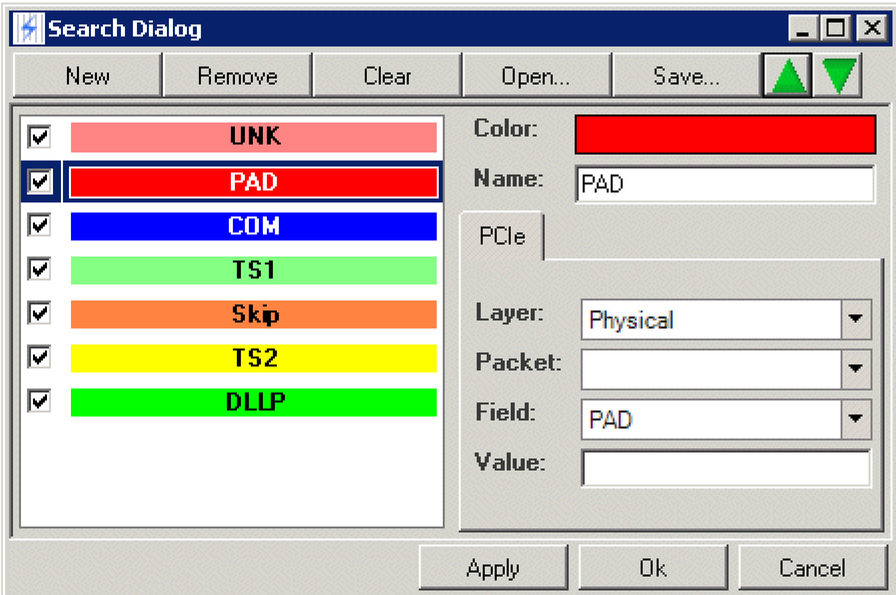
The New button adds a new item to the bottom of the search list:



Adding a New Item to the Search List

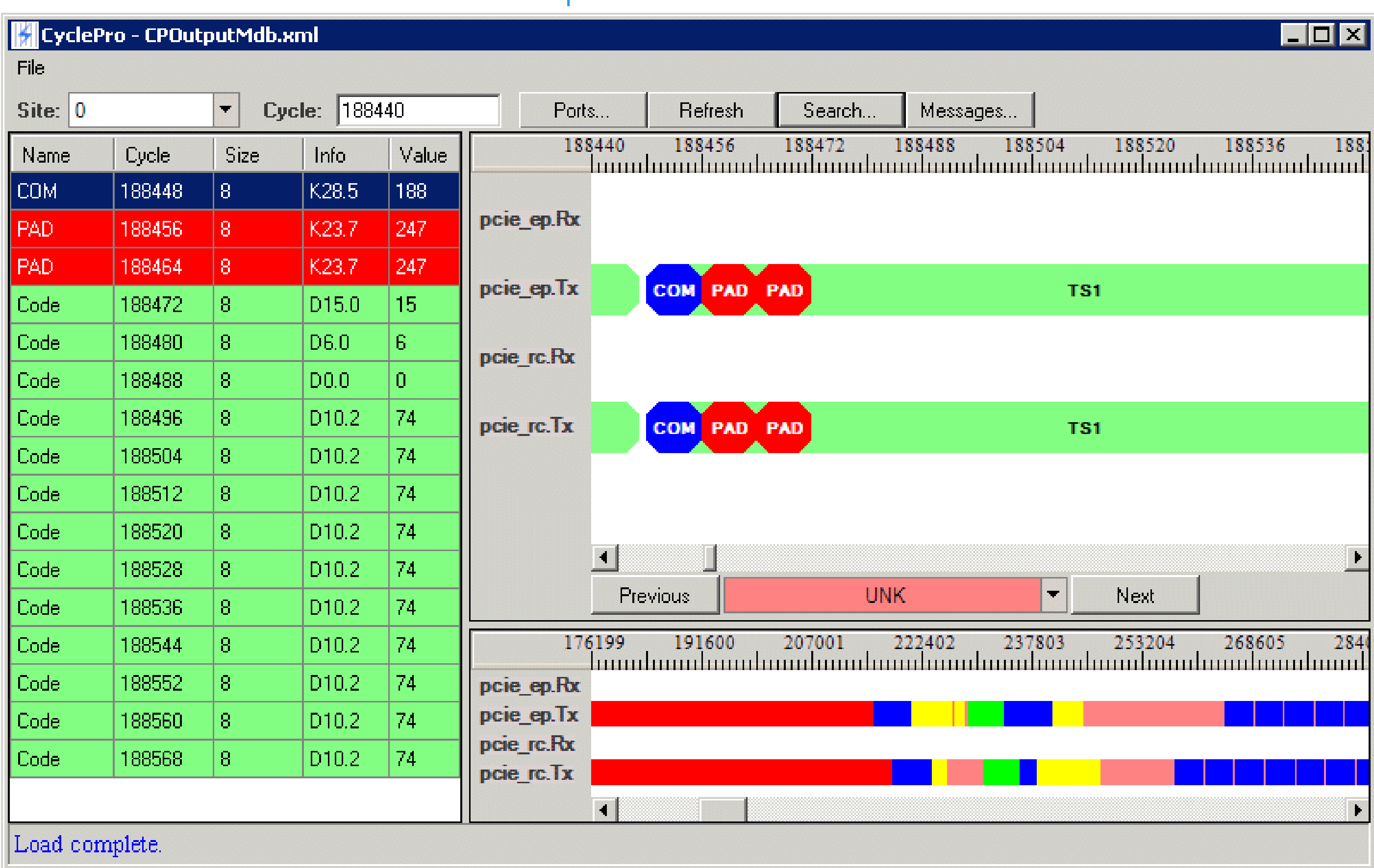
You now enter the values for the search yourself. You use the drop-down lists at the bottom right of the window to select the item to be searched for. You first choose what Layer: you want: Physical, Data Link, or Transaction. Your choice determines the options that appear in the other drop-down lists.

You can use any valid combination of Layer, Packet, Field, and Value. On valid combinations, see below. For example, suppose you want to color all physical packets that have the PAD field. You can set Layer to Physical and Field to PAD and leave Packet and Value blank. You also add a name to the Name field and select a nondefault color. You also move the PAD search above all other searches because you want its color applied before the colors for packets:



Changing Color and Priority in the Search List

This field is now colored red in the three panels:



PAD Color Red in Three Panels of CyclePro

The following are the valid combinations of Layer, Packet, Field, and Value:

- layer + packet - matches the given packet, including all fields within it
- layer + field - matches the given field within any packet and with any value
- layer + field + value - matches the given field with a given value in any packet
- layer + packet + field - matches a specific field within a specific packet with any value
- layer + packet + field + value - matches a specific field within a specific packet with a specific value

Note that Layer is always mandatory because the choice of layer determines the valid values for Packet and Field.

<	>

CycleProUsing.2.14

<	>

Using CyclePro : Using Search and Annotate to Analyze Data : Using the Search Dialog Buttons

Using the Search Dialog Buttons

The other buttons on the Search Dialog box are as follows:

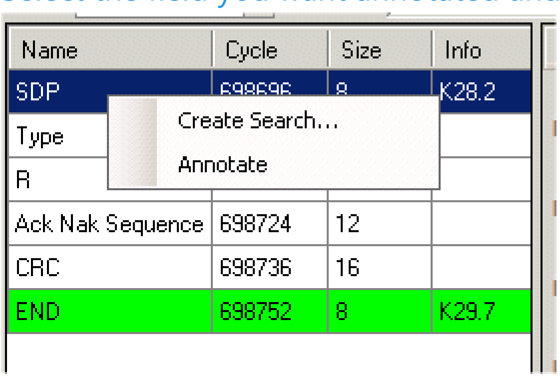
Remove	Remove the currently selected item from the search list. Instead of completely removing an item, you can uncheck the box next to it; this disables the item when you apply the search.
Clear	Clears all the items from the search list.
Open	Opens a previously saved search.
Save	Saves the current search list to an .xml file.

<	>

CycleProUsing.2.15

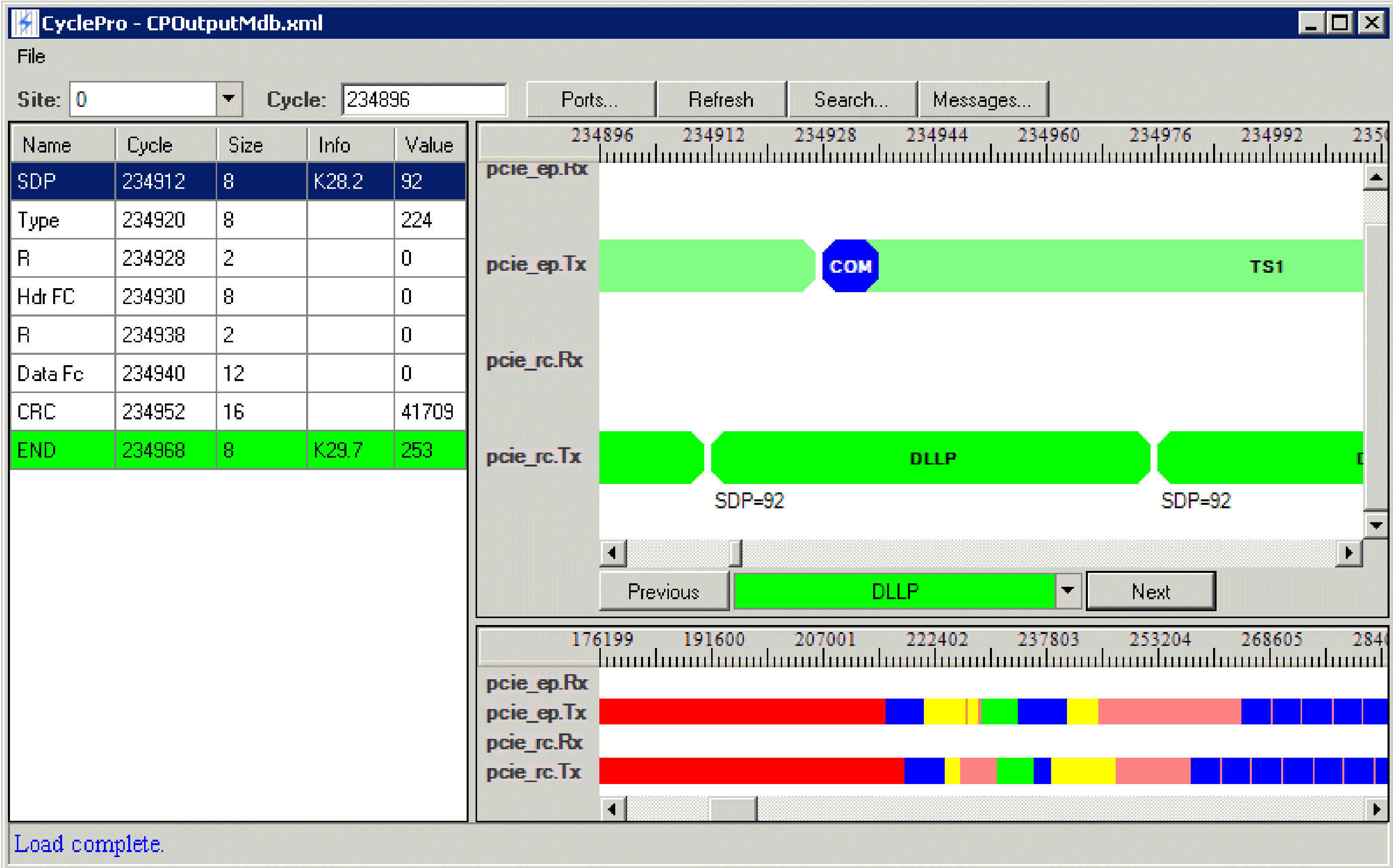
Annotating Field Data

You can also add field-level data values to the display using the Annotate capability. In the following example, you select a DLLP packet in the Packet Panel. The Field Panel displays the fields. Select the field you want annotated and added to the display: for example, SDP. Right-click:



Selecting a Field to Annotate

When you select Annotate, the field and its value are added to every packet that has that field:



CyclePro Display with Fields Annotated

The annotation is added to the DLLP packet, but not to the TS1 packet, because TS1 does not contain a SDP field.

< >